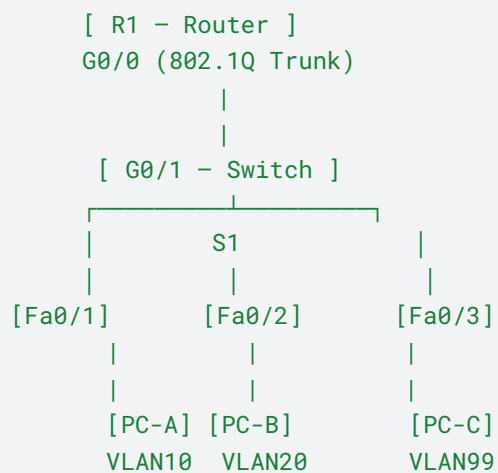
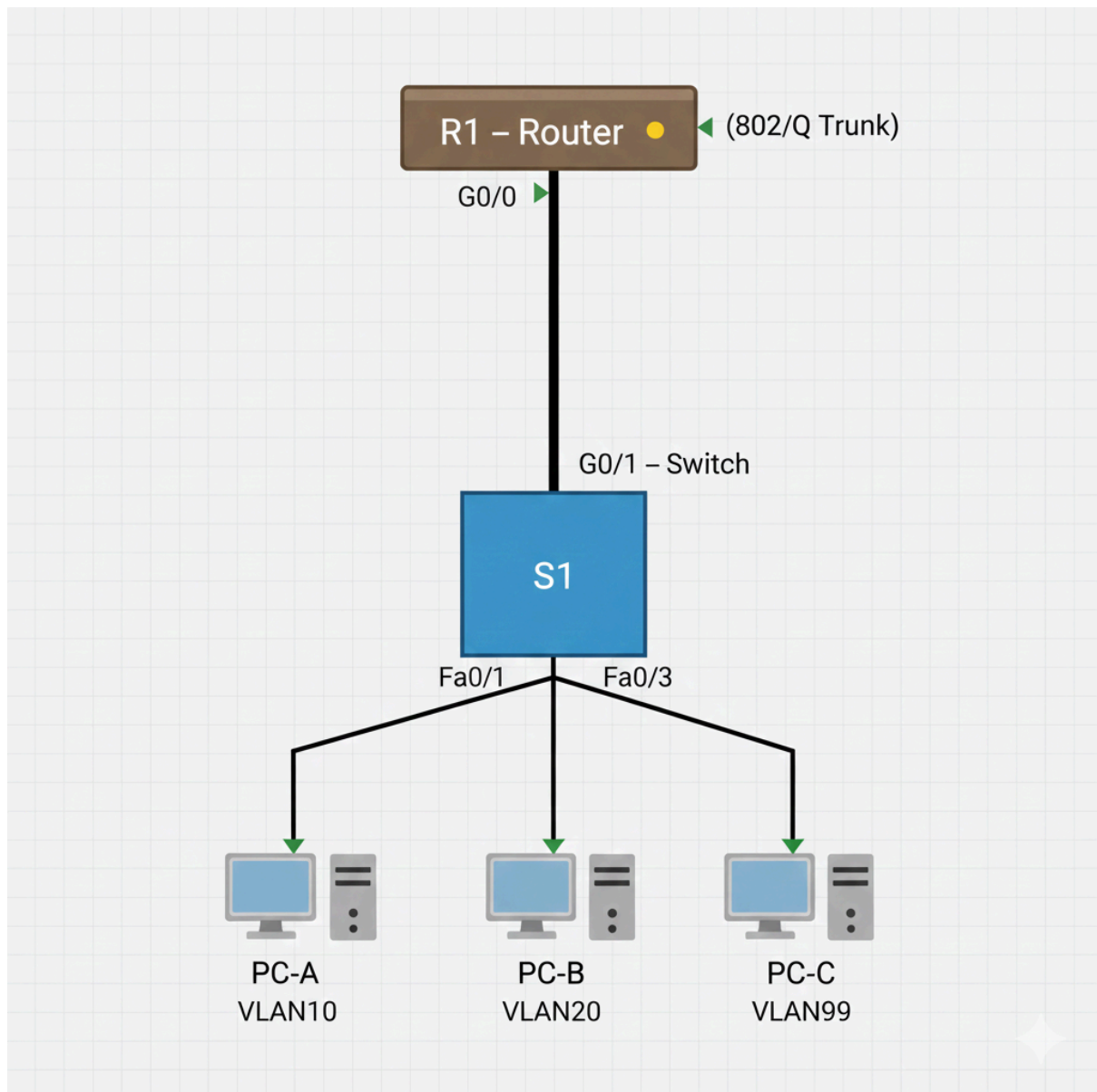

Lab 2: Secure & Scalable VLAN Design (Router-on-a-Stick)

TOPOLOGY DIAGRAM

None





ADDRESSING TABLE

Network Summary

- **Major Network:** 192.168.10.0/24
 - **Subnet Mask (/26):** 255.255.255.192
 - **Subnets Used:**
 - VLAN 10 → 192.168.10.0/26
 - VLAN 20 → 192.168.10.64/26
 - VLAN 99 → 192.168.10.128/26
-

Device Addressing

Device	Interface	VLAN	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	10	192.168.10.1	255.255.255.192	N/A
R1	G0/0.20	20	192.168.10.65	255.255.255.192	N/A
R1	G0/0.99	99	192.168.10.129	255.255.255.192	N/A
S1	VLAN 99	99	192.168.10.131	255.255.255.192	192.168.10.129
PC-A	NIC	10	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	NIC	20	192.168.10.70	255.255.255.192	192.168.10.65
PC-C	NIC	99	192.168.10.130	255.255.255.192	192.168.10.129

LAB OBJECTIVES

By completing this lab, you will be able to:

Part 1: Design and configure multiple VLANs on a switch **Part 2:** Implement Router-on-a-Stick for inter-VLAN routing **Part 3:** Apply subnetting using /26 networks **Part 4:** Isolate management and unused ports for security **Part 5:** Verify and troubleshoot Layer 2 and Layer 3 connectivity

LAB INSTRUCTION STEPS

PART 1: SWITCH CONFIGURATION (35 minutes)

Step 1: Basic Switch Setup

```
None
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
```

Step 2: Create VLANs

None

```
S1(config)# vlan 10
S1(config-vlan)# name USERS
S1(config-vlan)# exit

S1(config)# vlan 20
S1(config-vlan)# name SERVERS
S1(config-vlan)# exit

S1(config)# vlan 99
S1(config-vlan)# name MANAGEMENT
S1(config-vlan)# exit

S1(config)# vlan 999
S1(config-vlan)# name BLACKHOLE
S1(config-vlan)# exit
```

Step 3: Secure Unused Ports (Security First)

None

```
S1(config)# interface range fa0/4-24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# exit
```

Step 4: Assign Access Ports

None

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# no shutdown
```

```
S1(config-if)# exit

S1(config)# interface fa0/3
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 99
S1(config-if)# no shutdown
S1(config-if)# exit
```

Step 5: Configure Trunk to Router

```
None

S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,99
S1(config-if)# no shutdown
S1(config-if)# exit
```

Step 6: Configure Management SVI

```
None

S1(config)# interface vlan 99
S1(config-if)# ip address 192.168.10.131 255.255.255.192
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# ip default-gateway 192.168.10.129
S1(config)# end
S1# copy running-config startup-config
```

PART 2: ROUTER CONFIGURATION (30 minutes)

Step 7: Basic Router Setup

```
None

Router> enable
```

```
Router# configure terminal
Router(config)# hostname R1
```

Step 8: Enable Physical Interface

```
None
R1(config)# interface g0/0
R1(config-if)# no shutdown
R1(config-if)# exit
```

Step 9: Configure Router-on-a-Stick Subinterfaces

```
None
R1(config)# interface g0/0.10
R1(config-if)# encapsulation dot1Q 10
R1(config-if)# ip address 192.168.10.1 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.20
R1(config-if)# encapsulation dot1Q 20
R1(config-if)# ip address 192.168.10.65 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.99
R1(config-if)# encapsulation dot1Q 99
R1(config-if)# ip address 192.168.10.129 255.255.255.192
R1(config-if)# exit
```

```
None
R1(config)# end
R1# copy running-config startup-config
```

PART 3: END DEVICE CONFIGURATION (15 minutes)

Configure each PC manually.

PC-A (VLAN 10)

- IP: 192.168.10.10
- Mask: 255.255.255.192
- Gateway: 192.168.10.1

PC-B (VLAN 20)

- IP: 192.168.10.70
- Mask: 255.255.255.192
- Gateway: 192.168.10.65

PC-C (VLAN 99)

- IP: 192.168.10.130
 - Mask: 255.255.255.192
 - Gateway: 192.168.10.129
-

PART 4: CONNECTIVITY VERIFICATION (20 minutes)

Step 10: Verification Commands

From S1

```
None
show vlan brief
show interfaces trunk
```

From R1

```
None
show ip interface brief
```

From PCs

```
None
ping 192.168.10.1
ping 192.168.10.70
ping 192.168.10.131
```

Expected:

- PC-A ↔ PC-B ✓
 - PC-A ↔ S1 Management IP ✓
-

PART 5: TROUBLESHOOTING CHALLENGE (25 minutes)

Scenario A: Wrong VLAN Assignment

- Move PC-B to the wrong VLAN
 - Test connectivity
 - Identify failure point
 - Restore correct VLAN
-

Scenario B: Missing VLAN on Trunk

- Remove VLAN 99 from trunk
 - Test access to S1
 - Explain why management fails
 - Restore configuration
-

Scenario C: Wrong Gateway

- Change PC-A gateway
 - Observe ping failure
 - Use `ipconfig` and `ping`
 - Fix gateway
-

LAB DELIVERABLES

Worksheet 1: Configuration Check

Checkpoint	Command	Expected	Actual	Pass/Fail
VLANs exist	show vlan brief	VLANs 10,20,99		
Trunk active	show int trunk	VLANs allowed		
Router subifs	show ip int br	up/up		

Checkpoint	Command	Expected	Actual	Pass/Fail
Inter-VLAN ping	ping	Success		

Worksheet 2: Design Understanding

Answer briefly:

1. Why is VLAN 99 used for management?
 2. Why is VLAN 999 assigned to unused ports?
 3. What does 802.1Q tagging do?
 4. Why does each VLAN require a different subnet?
-

VERIFICATION COMMANDS CHECKLIST

Router

- show run
- show ip route
- show arp

Switch

- show vlan brief
- show interfaces trunk
- show mac address-table

PCs

- ipconfig
 - arp -a
 - ping
-

ASSESSMENT QUESTIONS

1. Which device performs inter-VLAN routing?
2. Why can PCs in different VLANs communicate?
3. What breaks if the trunk is removed?
4. Why is VLAN 1 avoided?
5. What security risk exists if unused ports remain active?

LAB COMPLETION CRITERIA

✓ All VLANs correctly created ✓ Router-on-a-Stick functioning ✓ Management VLAN isolated ✓ Unused ports secured ✓ All troubleshooting scenarios resolved
